C.S. MOTT CHILDREN'S HOSPITAL
MICHIGAN MEDICINE

ANTIBIOTIC TREATMENT GUIDELINES FOR PNEUMONIA IN PEDIATRIC PATIENTS

This guideline is designed to provide guidance for pneumonia in most children. Management of pneumonia in patients <3 months or in those who have underlying lung disease (e.g., cystic fibrosis; excluding asthma) or febrile neutropenia is beyond the scope of these guidelines. Other types of pneumonia such as Lemierre syndrome and atypical pneumonia in infants (pertussis, *C. trachomatis*) are also beyond the scope of these guidelines.

Indication	Empiric Therapy	Duration/Comments
Outpatient Community-Acquired Pneumonia Target pathogen: *S. pneumoniae*	1st line: **Amoxicillin*** 45 mg/kg/DOSE PO BID (max: 2 g/DOSE) Penicillin allergy: **Doxycycline** 2.2 mg/kg/DOSE PO BID (max: 100 mg/DOSE) OR **Clindamycin** 13 mg/kg/DOSE PO TID (max: 600 mg/DOSE)	• Duration: **5 days** • A **3-day** course of amoxicillin is encouraged for well-appearing outpatients (CAP-IT Trial, Bielicki et al., JAMA 2021) • For children <5 years, given predominance of viral pneumonia, consider supportive care only • High-dose amoxicillin, clindamycin, and doxycycline are preferred over oral cephalosporins, especially cefdinir, for the treatment of CAP caused by *S. pneumoniae* due to superior PK/PD parameters • Azithromycin resistance occurs in up to 40% of *S. pneumoniae*
Target pathogens: *M. pneumoniae* *C. pneumoniae*	Children ≥5 years with features of atypical pneumonia[1]: Consider **azithromycin** PO 10 mg/kg once on day 1 (max: 500 mg), followed by 5 mg/kg once daily x4 days (max: 250 mg/day) Azithromycin allergy/contraindication or to consolidate therapy for coverage of both *S. pneumoniae* and *Mycoplasma* in cases of diagnostic uncertainty: **Doxycycline** 2.2 mg/kg/dose PO BID (max: 100 mg/dose)	• The benefit of treatment for *Mycoplasma* is unclear. If epidemiology or clinical presentation is highly suggestive of atypical pneumonia,[1] or if not improving after 48 hours of typical pneumonia treatment, consider adding atypical coverage. Consider *Mycoplasma* PCR from oropharyngeal swab (Mayo test code MPRP); use shared decision-making for watchful waiting vs. adding empiric atypical coverage. Discontinue atypical coverage if PCR negative. • No additional atypical coverage needed if already using doxycycline • AAP RedBook now permits the use of doxycycline in any age group for courses ≤ 21 days

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Indication	Empiric Therapy	Step-Down Therapy	Duration/Comments
Non-severe / Uncomplicated Inpatient Community-Acquired Pneumonia **Excludes pneumonia with:** Loculated or moderate-large effusion, abscess, necrosis, bacteremia, septic shock requiring vasopressors, need for invasive mechanical ventilation Target pathogen: *S. pneumoniae* **Risk factors that require expanded antibacterial coverage:** Respiratory infection/colonization with MRSA, *P. aeruginosa*, or other organism resistant to non-severe CAP therapy in the past 12 months (see **Patients at Risk for Drug-Resistant Pneumonia**)	1st line: **Ampicillin*** 50 mg/kg/DOSE IV q6h (max: 2 g/DOSE) Low/High-risk penicillin allergy[2] OR failed high-dose amoxicillin[6]: **Ceftriaxone** 100 mg/kg once, then 50 mg/kg/DOSE IV q24h (max: 2 g/DOSE) Contraindication[4] to beta-lactams: **Levofloxacin***: *<5 years:* 10 mg/kg/DOSE IV/PO BID (max: 375 mg/DOSE) *≥5 years:* 10 mg/kg/DOSE IV/PO daily (max: 750 mg/DOSE)	1st line: **Amoxicillin*** 30 mg/kg/DOSE PO TID (max: 1 g/DOSE) Penicillin allergy: **Doxycycline** 2.2 mg/kg/DOSE PO BID (max: 100 mg/DOSE) OR **Clindamycin** 13 mg/kg/DOSE PO TID (max: 600 mg/DOSE) Failed high-dose amoxicillin[6]: **Levofloxacin***: *<5 years:* 10 mg/kg/DOSE PO BID (max: 375 mg/DOSE) *≥5 years:* 10 mg/kg/DOSE PO daily (max: 750 mg/DOSE)	• Duration: **5 days** (IV + oral) for patients who defervesce within 72 hours and have resolution of vital sign abnormalities by the time of antibiotic discontinuation • Patients with delayed response should discontinue therapy 48-72 hours after defervescence and when vital sign abnormalities have resolved • If vital signs remain abnormal >5 days but likely due to another etiology (e.g., bronchiolitis, asthma exacerbation), antibiotics may be discontinued at 5 days per clinical judgment • Suggest 5-7 days for infants <6 months based on clinical improvement • High-dose amoxicillin, clindamycin, doxycycline, and levofloxacin are preferred over oral cephalosporins, especially cefdinir, for the treatment of CAP caused by *S. pneumoniae* due to superior PK/PD parameters • Low-risk cephalosporin allergy[3]: Any penicillin may be utilized • Consider consultation to Beta-lactam Allergy Services while inpatient to assess whether amoxicillin can be trialed • Aspiration pneumonia • Empiric antibiotics are not indicated after an aspiration event or for aspiration pneumonitis. Aspiration pneumonitis typically resolves within 24-48 hours. • For patients who develop pneumonia following an aspiration event, anaerobic coverage is not necessary, unless lung abscess or necrosis are present. Prescribe routine pneumonia antibiotics. • Consider ID consult for significant prior antibiotics, no improvement with >48hrs guideline therapy, or anticipated prolonged antibiotics
Target pathogens: *M. pneumoniae* *C. pneumoniae*	Children ≥5 years with features of atypical pneumonia[1]: Add **azithromycin** PO 10 mg/kg once on day 1 (max: 500 mg), followed by 5 mg/kg once daily x 4 days (max: 250 mg/day) Azithromycin allergy/contraindication: **Doxycycline** 2.2mg/kg/dose POBID (max: 100 mg/dose)		• Discontinue if RPAN negative for atypical pathogens • If high suspicion for *Mycoplasma* infection, can send *Mycoplasma* testing using an oropharyngeal swab • No additional atypical coverage needed if already using levofloxacin • AAP RedBook now permits the use of doxycycline in any age group for courses ≤ 21 days

Setting	Empiric Therapy	Step-Down Therapy	Duration/Comments
Complicated Non-Severe Community-Acquired Pneumonia Complicated pneumonia includes loculated or moderate-large effusion, abscess, necrosis, and secondary bacteremia. Target pathogens: *S. pneumoniae* *S. aureus* *S. pyogenes* *H. influenzae* *M. catarrhalis* Anaerobes (if abscess or necrosis present) **Risk factors that require expanded antibacterial coverage:** Respiratory infection/colonization with MRSA, *P. aeruginosa*, or other organism resistant to non-severe CAP therapy in the past 12 months (see **Patients at Risk for Drug-Resistant Pneumonia**)	1st line: **Ampicillin-sulbactam*** 50 mg ampicillin/kg q6h (max: 2 g ampicillin/DOSE) + **MRSA nasal swab** Low/high-risk penicillin allergy[2]: **Ceftriaxone** 100 mg/kg once, then 50 mg/kg/DOSE IV q24h (max: 2 g/DOSE) + **MRSA nasal swab** High-risk cephalosporin allergy[3]/ contraindication[4] to beta-lactams, or low-risk allergy[2] to ceftriaxone/cefotaxime/ cefepime/cefpodoxime: **Levofloxacin***: *<5 years:* 10 mg/kg/DOSE IV/PO BID (max: 375 mg/DOSE) *≥5 years:* 10 mg/kg/DOSE IV/PO daily (max: 750 mg/DOSE) + **MRSA nasal swab** Beta lactam allergy with abscess or necrotizing pneumonia: Add **metronidazole*** 10 mg/kg/DOSE IV/PO q8h (max: 500 mg/DOSE) to **ceftriaxone** or **levofloxacin** If need for MRSA coverage: Add **Vancomycin***	1st line: **Amoxicillin-clavulanate***[5] 30 amoxicillin mg/kg/DOSE PO TID (max: 1 g amoxicillin/DOSE) Penicillin allergy: **Levofloxacin***: *<5 years:* 10 mg/kg/DOSE PO BID (max: 375 mg/DOSE) *≥5 years:* 10 mg/kg/DOSE PO daily (max: 750 mg/DOSE) If need for MRSA coverage: Add **TMP-SMX***[7] 6 mg TMP/kg/DOSE PO BID (max: 320 mg/DOSE) Penicillin allergy with abscess or necrotizing pneumonia: Add **metronidazole*** 10 mg/kg/DOSE PO q8h (max: 500 mg/DOSE) to **levofloxacin**	• Duration**:** • Consult ID, as longer duration is typically required for empyema, abscess, and bacteremia • Do not start empiric vancomycin for clinically stable patients with complicated pneumonia. MRSA nasal swab should be obtained to determine whether MRSA coverage is indicated. • If MRSA nasal swab is positive, discuss with ID consult whether MRSA coverage is needed

Severe Community-Acquired Pneumonia Severe Pneumonia is defined by septic shock requiring vasopressors or severe respiratory failure requiring invasive mechanical ventilation primarily attributed to bacterial pneumonia. Target pathogens: *S. pneumoniae* *S. aureus* *S. pyogenes* *H. influenzae* *M. catarrhalis* Anaerobes (if abscess or necrosis present) **Risk factors that require expanded antibacterial coverage:** • Infection/colonization with *P. aeruginosa* or other organism resistant to severe CAP therapy in the past 12 months • Severe pneumonia AND hospitalization >72hrs in the past 90 days (including current admission) with receipt of intravenous antibiotics • If present, see **Patients at Risk for Drug-Resistant Pneumonia**	1st line: **Ceftriaxone** 100 mg/kg once, then 50 mg/kg/DOSE IV q12h (max: 2 g/DOSE) + **Vancomycin*** High-risk cephalosporin allergy[3]/ contraindication[4] to beta-lactams or low-risk allergy[2] to ceftriaxone/cefotaxime/ cefepime/cefpodoxime: **Levofloxacin***: *<5 years:* 10 mg/kg/DOSE IV/PO BID (max: 375 mg/DOSE) *≥5 years:* 10 mg/kg/DOSE IV/PO daily (max: 750 mg/DOSE) + **Vancomycin*** Abscess or necrotizing pneumonia: Substitute **ampicillin-sulbactam** 50 mg ampicillin/kg/DOSE IV q6h (max: 2 g ampicillin/DOSE) for **ceftriaxone** -OR- Add **metronidazole*** 10 mg/kg/DOSE IV/PO q8h (max: 500 mg/DOSE) to **levofloxacin**	1st line: **Amoxicillin-clavulanate***[5] 30 amoxicillin mg/kg/DOSE PO TID (max: 1 g amoxicillin/DOSE) Penicillin allergy: **Levofloxacin***: *<5 years:* 10 mg/kg/DOSE PO BID (max: 375 mg/DOSE) *≥5 years:* 10 mg/kg/DOSE PO daily (max: 750 mg/DOSE) If need for MRSA coverage: Add **TMP-SMX***[7] 6 mg TMP/kg/DOSE PO BID (max: 320 mg/DOSE) Penicillin allergy with abscess or necrotizing pneumonia: Add **metronidazole*** 10 mg/kg/DOSE PO q8h (max: 500 mg/DOSE) to **levofloxacin**	• Duration: **5 days** (IV + oral) for patients who defervesce within 72 hours and have resolution of vital sign abnormalities by the time of antibiotic discontinuation • Patients with delayed response should discontinue therapy 48-72 hours after defervescence and when vital sign abnormalities have resolved • **7 days** (IV + oral) for patients with *S. aureus*, *P. aeruginosa* or other NLFGNR[8] infection • Consult ID for complicated pneumonia as longer duration is typically required for empyema, abscess, and bacteremia • Obtain MRSA nasal swab when starting vancomycin and discontinue vancomycin if negative and no concurrent gram-positive bacteremia. TMP-SMX is not needed for step-down if vancomycin is discontinued. • Tailor therapy to culture results; if cultures negative, de-escalate to **ampicillin-sulbactam** or **amoxicillin-clavulanate** • High-dose amoxicillin, clindamycin, doxycycline, and levofloxacin are preferred over oral cephalosporins, especially cefdinir, for the treatment of CAP caused by *S. pneumoniae* due to superior PK/PD parameters
See below for severe atypical pneumonia			

Severe / Complicated Atypical Pneumonia Target pathogens: *M. pneumoniae* *C. pneumoniae* *Legionella* spp.	Empiric coverage for children ≥5 years with features of atypical pneumonia[1]: Add **azithromycin** IV/PO 10 mg/kg once on day 1 (max: 500 mg), followed by 5 mg/kg once daily x 4 days (max: 250 mg/day) Azithromycin allergy/contraindication and no concern for *Legionella*: **Doxycycline** 2.2 mg/kg/dose IV/PO BID (max: 100 mg/dose) Severe documented *Mycoplasma* infection: **Doxycycline** 2.2 mg/kg/dose IV/PO BID (max: 100 mg/dose) Documented or high clinical suspicion for *Legionella* infection: **Azithromycin** 10 mg/kg IV/PO q24h (max: 500 mg) Azithromycin allergy/contraindication and *Legionella* infection: **Levofloxacin***: *<5 years:* 10 mg/kg/DOSE IV/PO BID (max: 375 mg/DOSE) *≥5 years:* 10 mg/kg/DOSE IV/PO daily (max: 750 mg/DOSE)	• Consider discontinuing empiric coverage if RPAN is negative for atypical pathogens • If high suspicion for *Mycoplasma* infection, can send *Mycoplasma* testing using an oropharyngeal swab • No additional atypical coverage needed if already using levofloxacin • If suspicion for *Legionella* infection, send urine *Legionella* antigen and, if possible, respiratory *Legionella* PCR • In patients with severe documented *Mycoplasma*, doxycycline is preferred for treatment due to rising macrolide resistance • AAP RedBook now permits the use of doxycycline in any age group for courses ≤ 21 days

Setting	Empiric Therapy	Step-Down Therapy	Duration/Comments
Patients at Risk for Drug-Resistant Pneumonia Includes: Ventilator-Associated Pneumonia Ventilator-Associated Tracheobronchitis Patients with non-severe pneumonia and respiratory infection/colonization with MRSA, *P. aeruginosa*, or other organism resistant to non-severe CAP therapy in the past 12 months Patients with severe pneumonia and infection/colonization with *P. aeruginosa*, or other organism resistant to severe CAP therapy in the past 12 months Patients with severe pneumonia and hospitalization >72hrs in the past 90 days (including current admission) with receipt of intravenous antibiotics Target pathogens: *S. pneumoniae* *S. aureus* *P. aeruginosa* *H. influenzae* *M. catarrhalis*	**Empiric therapy should be guided by previous culture data.** Patients with non-severe CAP do NOT require cefepime or vancomycin unless dictated by cultures. 1st line (unless alternative regimen indicated by previous culture data as above): **Cefepime*** 50 mg/kg/DOSE IV extended infusion q8h (max 2 g/DOSE) + **Vancomycin*** High-risk cephalosporin allergy[3]/ contraindication[4] to beta-lactams or low-risk allergy[2] to ceftriaxone/cefotaxime/ cefepime/cefpodoxime: **Levofloxacin***: *<5 years:* 10 mg/kg/DOSE IV/PO BID (max: 375 mg/DOSE) *≥5 years:* 10 mg/kg/DOSE IV/PO daily (max: 750 mg/DOSE) + **Vancomycin*** Abscess or necrotizing pneumonia: Add **metronidazole*** 10 mg/kg/DOSE IV/PO q8h (max: 500 mg/DOSE)	Negative culture results: Use step-down therapy for non-severe or severe CAP as appropriate Positive culture results: Tailor therapy to culture results	• Duration: **7 days** for ventilator-associated pneumonia (VAP) **5 days** for ventilator-associated tracheobronchitis (VAT), if patient has new purulent endotracheal secretions without chest x-ray infiltrate plus 1) fever/hypothermia and 2) leukocytosis/leukopenia or elevated/rising inflammatory markers. Consider not treating with systemic antibiotics for patients with less severe signs/symptoms Non-VAP: See applicable sections for Non-severe or Severe pneumonia • Obtain MRSA nasal swab when starting vancomycin and discontinue vancomycin if negative and no concurrent bacteremia • If patient has a negative MRSA nasal swab within the past 14 days and is not in septic shock, do not start empiric vancomycin • Patients with mild-moderate VAP/VAT may be started on alternative regimens to better target known organisms, or if tolerating PO and suitable oral alternatives exist

Setting	Empiric Therapy	Duration/Comments
Emergency Department	Anticipated discharge to home: • Prescribe antibiotics per Outpatient recommendations • If failed high-dose amoxicillin[6] for typical bacterial pneumonia, use levofloxacin Hypoxemic and/or not tolerating PO but not progressing toward fluid-refractory septic shock or intubation; no loculated or moderate-large effusion, abscess, or necrosis: • Begin empiric therapy per Non-severe / Uncomplicated recommendations • Transition to Outpatient empiric therapy if able to discharge Progressing toward fluid-refractory septic shock/intubation, or pneumonia with loculated or moderate-large effusion, abscess, or necrosis: • Begin empiric therapy per Severe / Complicated recommendations • If signs/symptoms of sepsis resolve and patient does not have bacteremia, empyema, abscess, or necrosis, transition to Non-severe recommendations	• Duration for outpatient therapy: See Outpatient section above

*Renal adjustment may be necessary. See Pediatric Antimicrobial Dosing Guidelines.

[1]Atypical pneumonia is characterized by slow progression of symptoms (over 3-5 days); typical signs/symptoms include, but are not limited to: malaise, sore throat, headache, cough, low-grade fever, and non-focal auscultatory and chest x-ray findings

[2]Low-risk allergies include: pruritus without rash, urticaria/hives with no other symptoms, remote (>10 years) unknown reaction, patient denies allergy but is on record, mild to severe rash with no other symptoms (mild rash: non-urticarial rash that resolves without medical intervention). See β-lactam allergy evaluation and empiric guidance for further information.

[3]High-risk allergies include: anaphylaxis, respiratory symptoms, angioedema/swelling, cardiovascular symptoms (syncope/passing out, arrhythmia). See β-lactam allergy evaluation and empiric guidance for further information.

[4]Severe non-IgE allergy: organ damage (kidney, liver), drug-induced immune-mediated anemia/thrombocytopenia/leukopenia, rash with mucosal lesions (Stevens Johnson Syndrome/Toxic Epidermal Necrosis), rash with pustules (acute generalized exanthematous pustulosis), rash with eosinophils and organ injury (DRESS – drug rash eosinophilia and systemic symptoms), rash with joint pain, fever, and myalgia (Serum Sickness). See β-lactam allergy evaluation and empiric guidance for further information.

[5]Use amoxicillin-clavulanate ES (600 mg/42.9 mg/5 mL) oral suspension to limit the risk of diarrhea associated with high doses of clavulanate. For patients >30 kg who prefer tablets, can use amoxicillin-clavulanate 875-125 mg/dose Q8H.

[6]Refers to patients who were compliant with, and tolerated, oral high-dose amoxicillin for >48 hours

[7]TMP-SMX: trimethoprim-sulfamethoxazole

[8]NLFGNR: non-lactose-fermenting gram-negative rods, which include *Pseudomonas aeruginosa*, *Stenotrophomonas maltophilia*, *Acinetobacter* spp., and *Achromobacter* spp.

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Revision History	
Date	**Revision**
03/2021	Updated vancomycin goals
12/2021	Major revision: decreased treatment duration, substituted ceftriaxone for clindamycin inpatient, defined severe/complicated pneumonia, added section on drug-resistant pneumonia
01/2025	Removed recommendation for broader empiric therapy in underimmunized patients; added separate section for complicated non-severe pneumonia; revised beta lactam allergy recommendations; added *Mycoplasma* testing guidance

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source